Implementing cooperative link diagnostic

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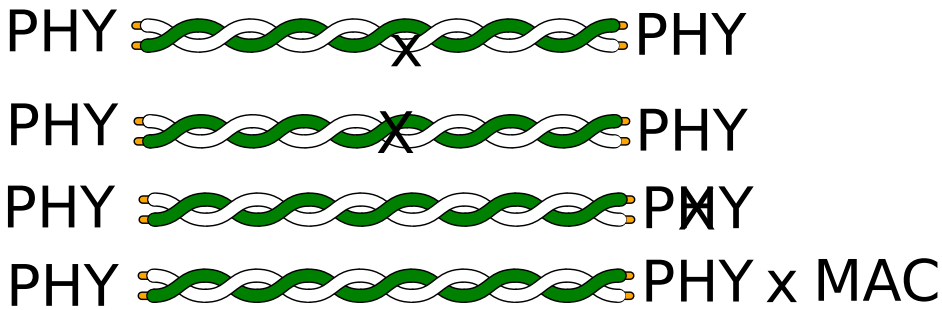
What could possibly go wrong?!

- Industrial and building automation challenges:
  - 10Base-T1L: 1000-2000 meter copper twisted pair.
  - Remote devices are hard to access
  - Mechanical and thermal stress
  - In field repair attempts
  - Animals

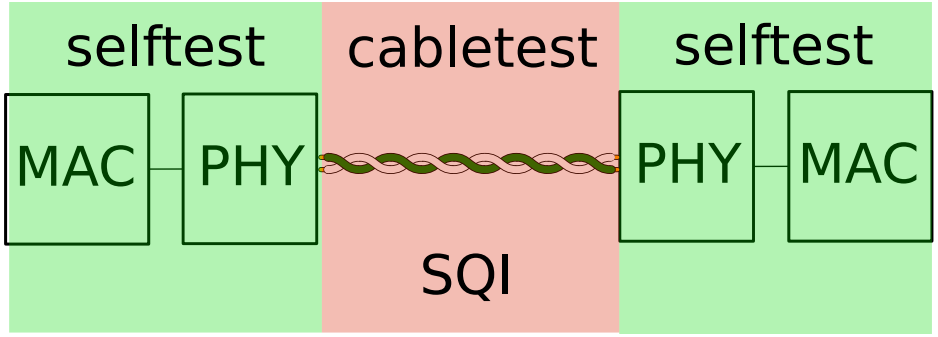
Variants of “broken”

- One of cables of twisted pairs is open
- Complete twisted pair is open.
- Short withing pair
- Short between pairs
- Short to ground
- … many more …



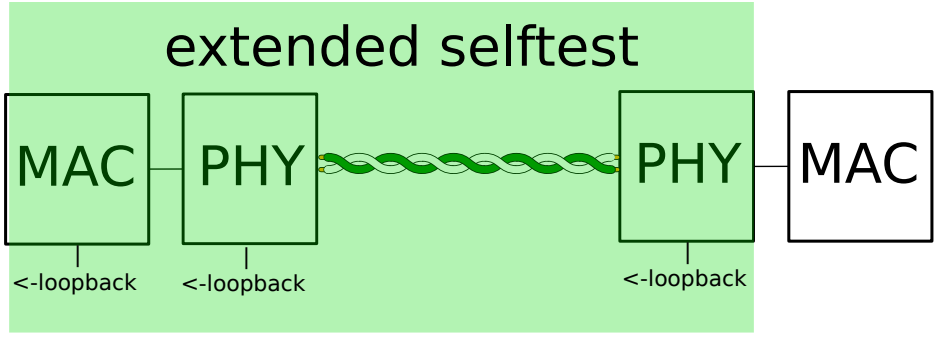
Currently available diagnostic

- selftest:
  - ethtool -t eth0
- cabletest
  - ethetool –cable-test eth0
- SQI
  - ethtool eth0 | grep -i sqi



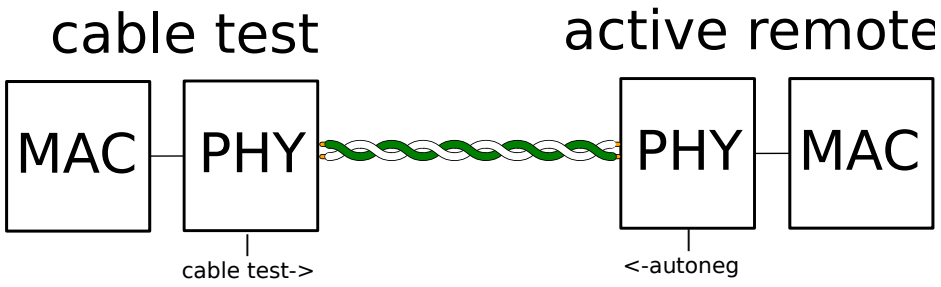
Extending selftest

- It is possible to extend selftest. Local and remote system should cooperate with each other.
- remote PHY should be able to enable external loopback on request
- How to communicate with remote system?



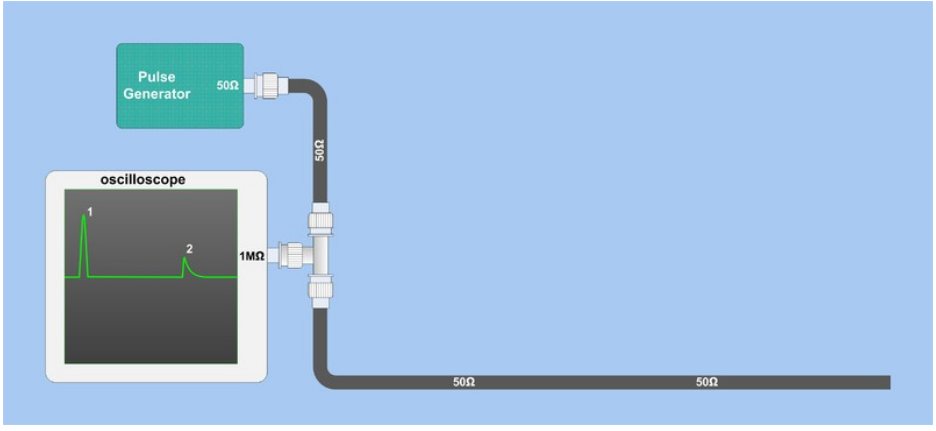
Optimizing cable test

- No problems If cable is completely damaged or remote system is off
- If remote system is active, cable test is disturbed by autoneg FLPs.
- We need to understand more about cable testing and autoneg to optimize it.

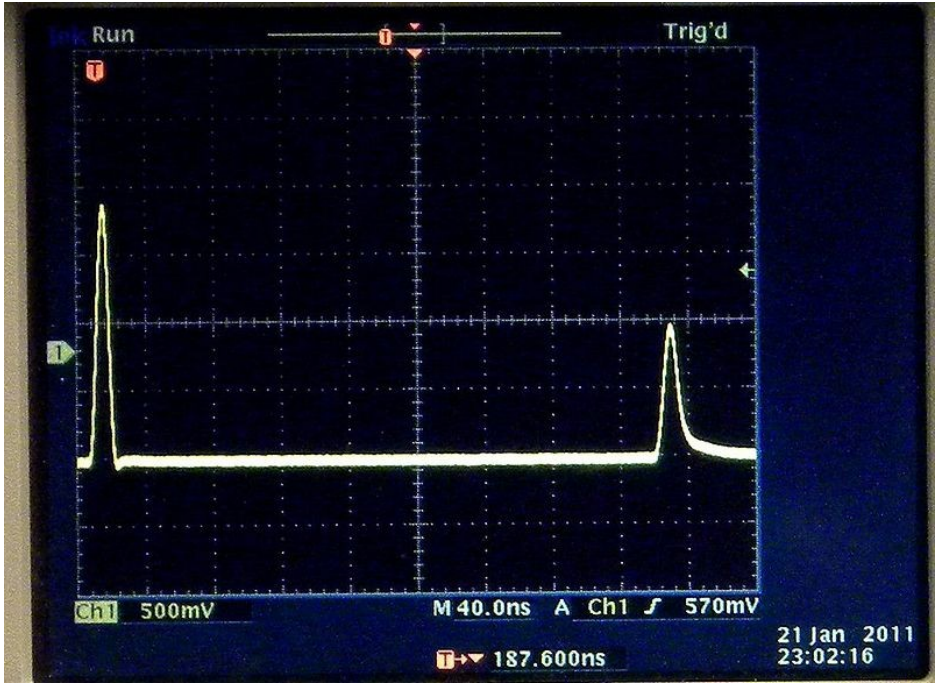


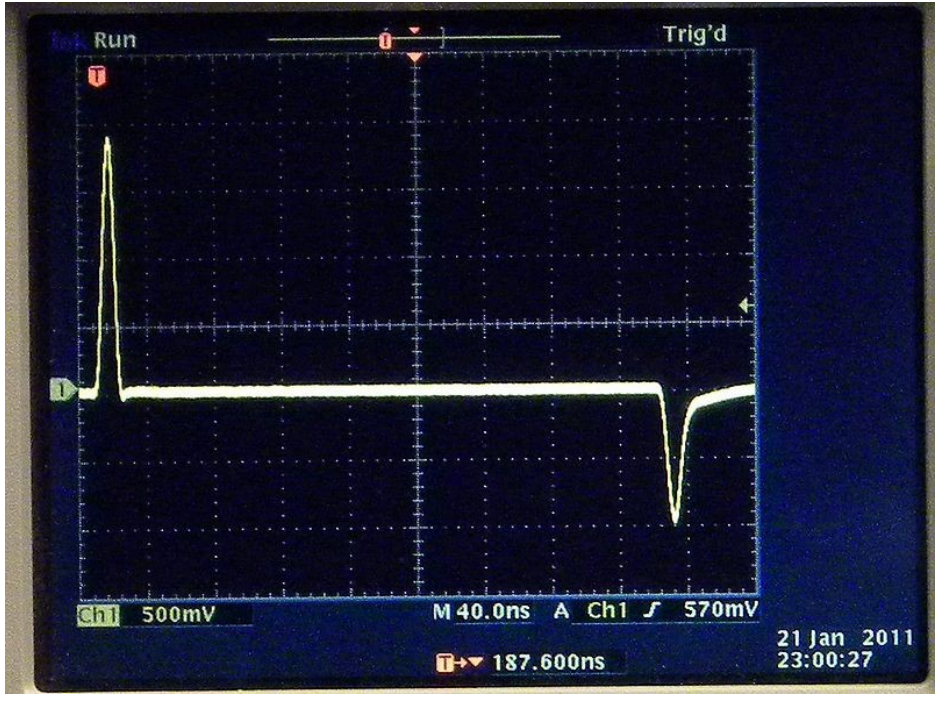
Cable diagnostic with TDR

- Time-Domain Reflectometery
- PHY is pulse generator and oscilloscope
- Precision depends on sampling rate
- Usual PHYs can get about ~80cm precision.

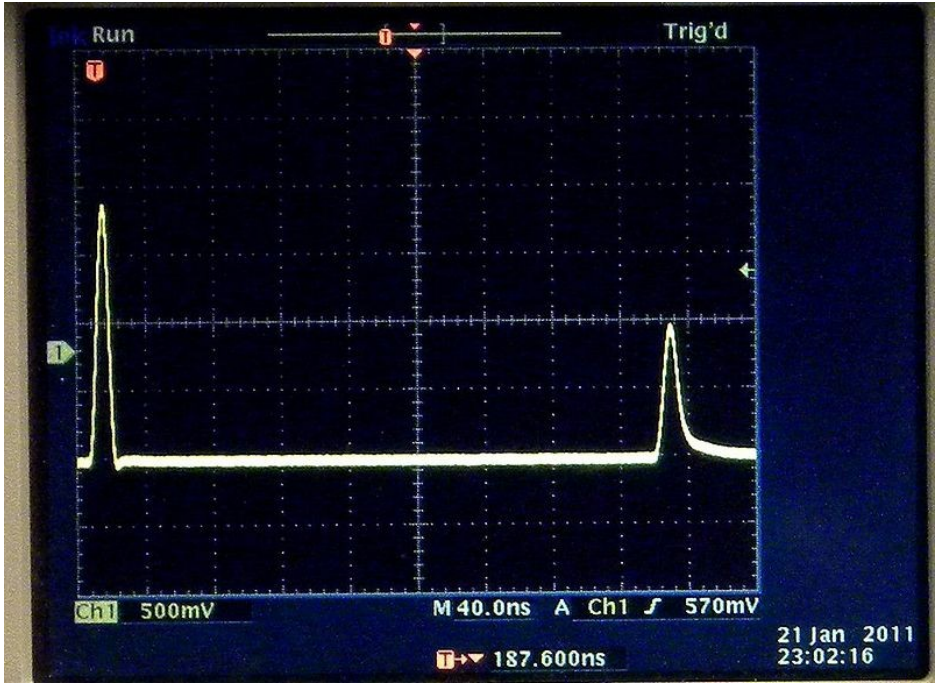


TDR: Open/Short recognition





TDR vs Autoneg



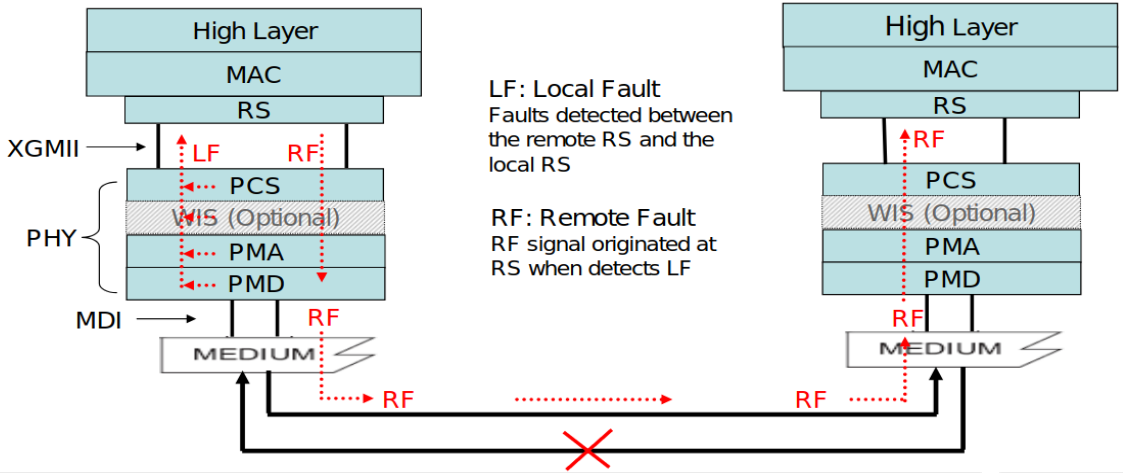
Variants of “broken”

- No signal
  - Good for cable testing. No interference from link partner.
  - No special handling is needed.
- Some signal is present
  - Cable testing is disturbed by autoneg pulses, attempt to create link or manually configured link.
  - Proposal: For the autoneg, use “remote fault” flag to request a “silence window” on the wire.

It is not a new idea

PHY Monitoring in 802.3ae

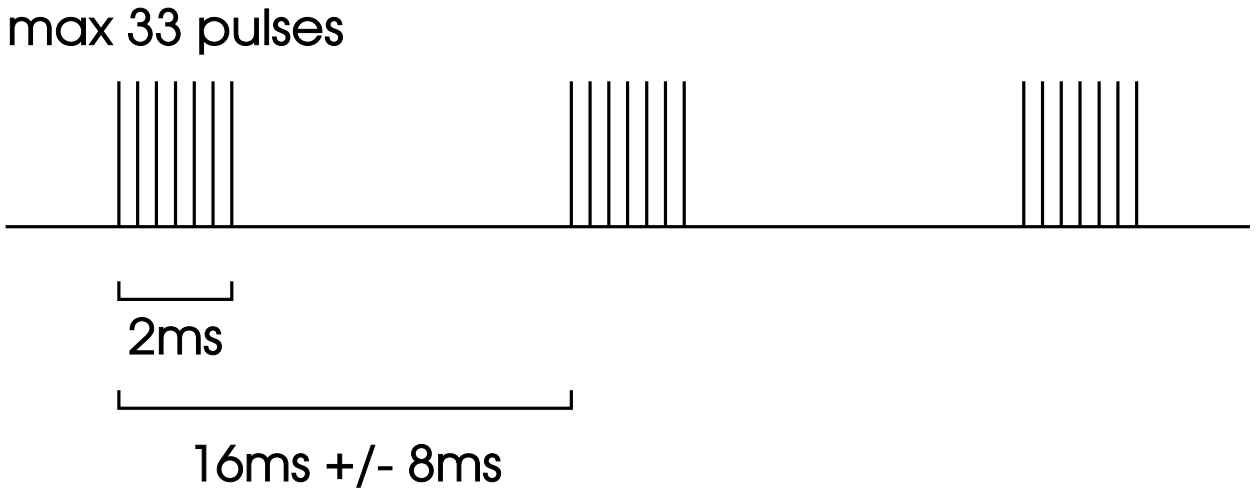
- **There is a Link Fault Signaling mechanism in 10GE**
  - Reside in Reconciliation Sublayer (RS)
  - To monitor link status between local RS & remote RS and perform link status notification
  - Sublayers within the PHY are required to detect faults that render a link unreliable



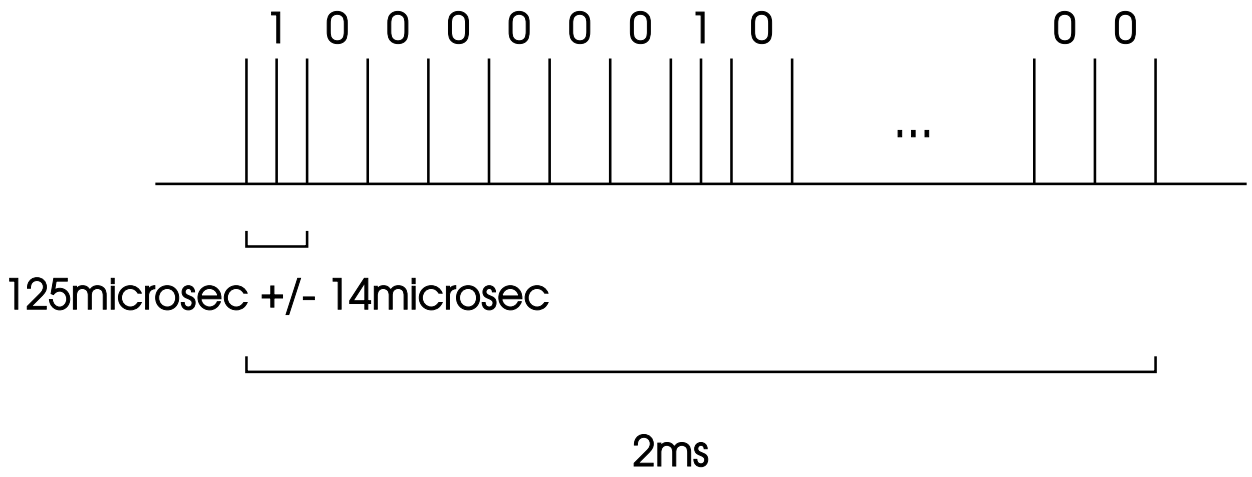
How this works

- Autonegotiation pulses run at lower frequency and less affected by partially damaged twisted pair.
- IEEE 802.3 defines standard to communicate by using autoneg pulses.
- How about to run IP over autoneg? ;)

Autoneg FLP (Fast link pulses)



FLP: Link code word



Link code word

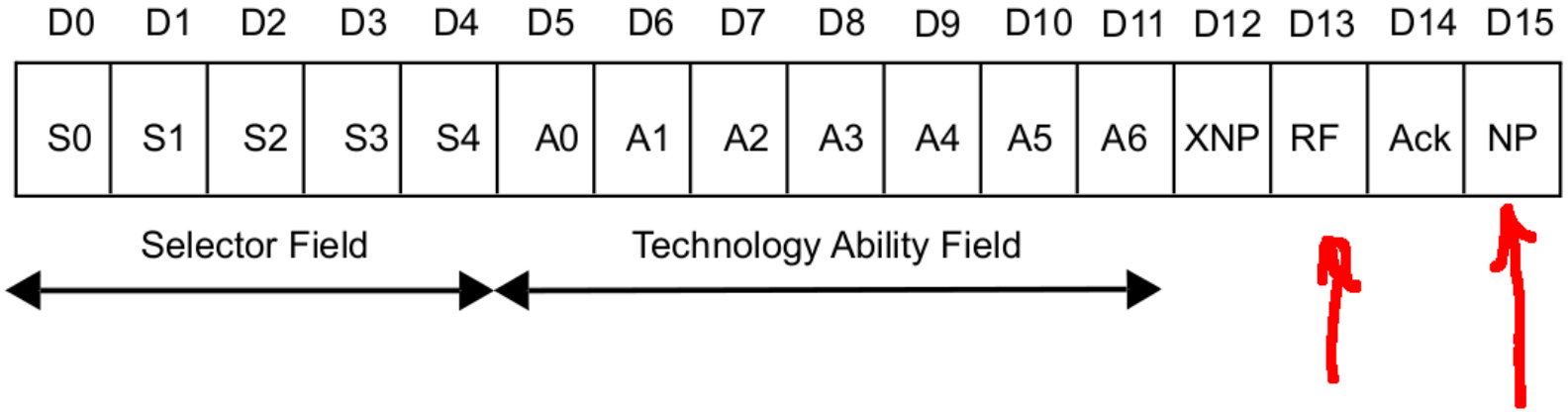


Figure 28–7—Base Page encoding

Link code word

- We can use “remote fault” (RF) flag to notify about some error state.
- If we want to be more specific we need read/write own pages and set “next page” (NP) flag.
- I can image to have following requests:
  - Silent autoneg for X amount of time
  - Enable remote loopback on the PHY or MAC
  - ...More ideas?...

Other pages

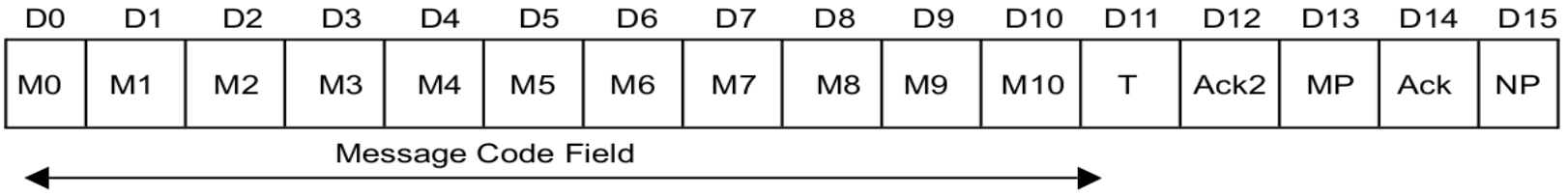


Figure 28–11—Message Page encoding

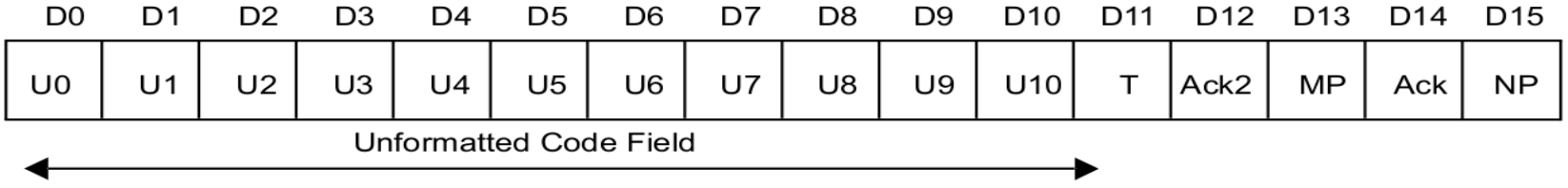


Figure 28–12—Unformatted Page encoding

Extended pages

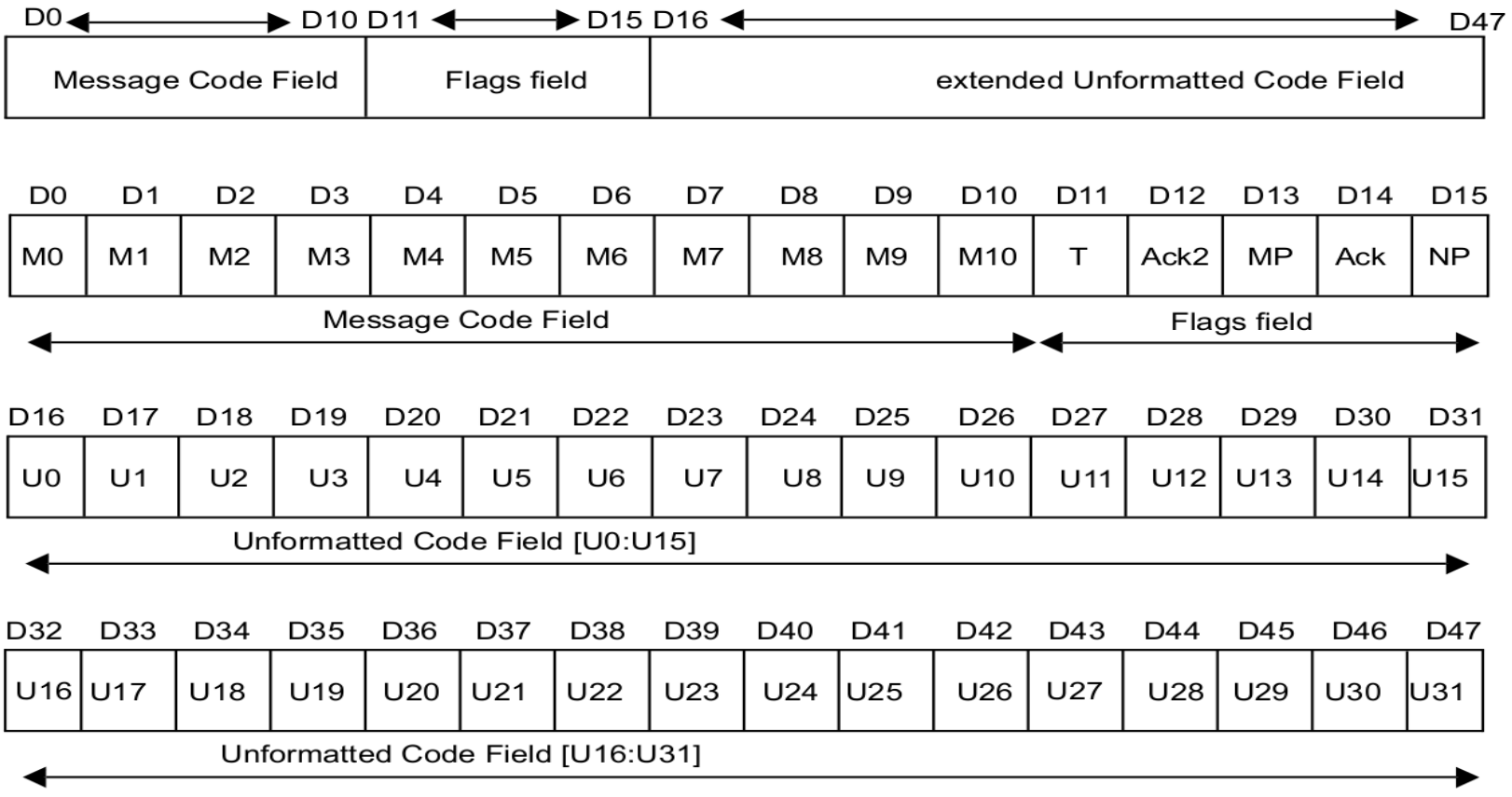


Figure 28–13—Extended Message Page encoding

Kernel integration

- Initial patch set (v1) was exporting every thing to the user space.
- For v2 patches I’ll need to move RF handling to the kernel.

Patch v1: add remote fault support

- https://lore.kernel.org/all/20220608093403.3999446-1-o.rempel@pengutronix.de/

Thank you!

Questions?